DAY 7 | EXCEL

My Data Analyst Journey

Random, Rounding & LARGE / SMALL

Day 7 of learning Excel in public.

F9 Shift F3

Swipe to learn with me >

Random Numbers

RAND, RANDBETWEEN, RANDARRAY

`=RAND()`	Decimal from 0 to 1
`=RANDBETWEEN(1,100)`	Whole number 1 to 100
`=RANDARRAY(10,3,0,100,TRUE)`	10 x 3 whole numbers
`=RANDARRAY()`	Decimal 0 to 1, spills

RANDBETWEEN(bottom, top): the smallest number comes first. RANDARRAY(rows, columns, min, max, whole_number) works in Microsoft 365 and Excel 2021 or newer.

Why Do They Keep Changing?

Random functions are volatile

Volatile

RAND, RANDBETWEEN and RANDARRAY get new numbers every time the sheet recalculates.

Recalculate

Press F9 to get a fresh set of random numbers.

Freeze Them

Copy > Ctrl+Alt+V > V > Enter. The numbers become fixed values.

Other Volatile Ones

TODAY() and NOW() also refresh on every recalculation.

ROUND

=ROUND(number, num_digits)

`=ROUND(186.74648,3)`	186.746
`=ROUND(186.74648,2)`	186.75
`=ROUND(186.74648,0)`	187
`=ROUND(186.74648,-1)`	190
`=ROUND(186.74648,-2)`	200

Positive digits = decimals. 0 = whole number. Negative digits round to the left (tens, hundreds). The digit 5 or above rounds up.

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ROUNDUP & ROUNDDOWN

Force the direction

```
=ROUNDUP(186.74648,3)
```
186.747

```
=ROUNDUP(186.74648,0)
```
187

```
=ROUNDDOWN(186.74648,3)
```
186.746

```
=ROUNDDOWN(186.74648,1)
```
186.7

```
=ROUNDDOWN(186.74648,0)
```
186

ROUNDUP always moves away from zero and ROUNDDOWN toward zero. Unlike ROUND, there is no "5 rule".

INT vs TRUNC

Remove the decimals

`=INT(186.74648)`	186
`=TRUNC(186.74648)`	186
`=TRUNC(186.74648,1)`	186.7
`=INT(-5.5)`	-6
`=TRUNC(-5.5)`	-5

INT rounds down to the next lower whole number. TRUNC simply cuts off the decimals. They differ for negative numbers. TRUNC can also keep some decimals.

LARGE & SMALL

=LARGE(range, k) and =SMALL(range, k)

`=LARGE(B2:B11,1)`	**821 (biggest)**
`=LARGE(B2:B11,2)`	**813**
`=LARGE(B2:B11,3)`	**528**
`=SMALL(B2:B11,1)`	**128 (smallest)**
`=SMALL(B2:B11,2)`	**206**

Data: 821, 486, 528, 128, 422, 206, 342, 296, 813, 330. k = which position you want. LARGE(range,1) is the same as MAX, and SMALL(range,1) is the same as MIN.

Top N Tricks

Use LARGE and SMALL smartly

Second Highest

=LARGE(B2:B11,2). A very common interview question.

Top 3 Total

=SUM(LARGE(B2:B11,{1,2,3})) adds the three biggest numbers.

Highlight Top 3

Conditional Formatting > Top/Bottom Rules > Top 10 Items > change 10 to 3.

Errors

If k is bigger than the number of values you get #NUM!.

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Day 7 Shortcuts

Calculate and control

Recalculate now	F9
Calculate this sheet only	Shift + F9
Recalculate everything	Ctrl + Alt + F9
Freeze as values	Ctrl + Alt + V + V
Insert Function box	Shift + F3
Function Arguments (after =ROUND( )	Ctrl + A
Increase decimal	Alt + H + 0
Decrease decimal	Alt + H + 9

Interview Questions

Short and sharp answers

ROUND vs decimal buttons?

Alt+H+9 only changes how the number looks. ROUND changes the real value.

INT vs TRUNC?

Same for positives. For -5.5, INT gives -6 and TRUNC gives -5.

Why do random numbers change?

They are volatile. Press F9 or any edit recalculates them. Paste Values to freeze.

Find the second highest?

=LARGE(range,2). The second lowest is =SMALL(range,2).

More Rounding Tools

Round to a multiple

`=MROUND(17,5)`	**15 (nearest multiple)**
`=CEILING(17,5)`	**20 (up to multiple)**
`=FLOOR(17,5)`	**15 (down to multiple)**
`=EVEN(5)`	**6**
`=ODD(4)`	**5**

Useful for prices, time slots and packaging sizes.

Day 7 Cheat Sheet

All functions in one place

Random

RAND, RANDBETWEEN, RANDARRAY

Round

ROUND, ROUNDUP, ROUNDDOWN

Round to Multiple

MROUND, CEILING, FLOOR, EVEN, ODD

Cut Decimals

INT, TRUNC

Position in a List

LARGE, SMALL, MAX, MIN, RANK

Day 7
Done!

Random numbers, rounding and
LARGE / SMALL.

Can you find the 2nd highest value?
Tell me in the comments.

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